

Implementation: Interactive Visual System



Milestone 4: From Prototype to Working Visualization for Jersey City Citi Bike Analysis

Group: 11

GitHub Repository: <https://github.com/tsaichen10/jc-citibike-vis>

The system follows the **Information Seeking Mantra**: *Overview first, zoom and filter, then details-on-demand*. The interface is structured into three coordinated zones:

- **Center:** A geographic map visualizing spatial flow.
- **Bottom:** A time-series chart for temporal trends and interactive filtering.
- **Side:** A dynamic ranking chart for station-level performance and connectivity analysis.

Requirement 1: Views & Encodings

The implementation successfully encodes the data abstraction through coordinated visual marks and channels:

- **Spatial View (Map):** Station locations are encoded as **points** (position). Flows are implemented as **curved arcs**, where **line width** and **opacity** represent trip volume (magnitude), facilitating the identification of major transit corridors.
- **Temporal View (Line Chart):** Uses **position (vertical)** for trip counts and **position (horizontal)** for the 24-hour cycle. **Color hue** (Blue for Members, Orange for Casual) effectively distinguishes user demographics.
- **Ranking View (Bar Chart):** Utilizes **length** and **aligned vertical position**—the most effective channels for quantitative comparison—to rank station popularity and flow balance.

Requirement 2: Interactions that Matter

We have implemented interactive features that enable direct task completion without ornamental clutter:

- **Brushing & Filtering:** Users can drag a selection box across the time-series chart. This **brush** interaction triggers an instant update of the Map and Bar Chart, allowing users to isolate specific periods like the morning rush hour.
- **Details-on-Demand:** Hovering over any visual mark (stations, arcs, or line nodes) reveals precise statistics via **tooltips**. Clicking a station expands a **detailed info panel** with balance metrics and peak usage data.
- **Search & Auto-Complete:** A search bar allows users to locate specific stations instantly, automatically centering the map and highlighting the station across all views.
- **Temporal Animation:** A "Play" button provides an automated walkthrough of the 24-hour cycle, visualizing the dynamic pulse of the city's bike flow.

Requirement 3: State & Coordination

Centralized state management ensures the system remains synchronized:

- **Linked Views:** Selecting a station on the map or ranking list updates the entire dashboard. The line chart adapts to show that station's specific temporal profile, while the map filters to its "1-hop neighborhood" (direct connections).
- **Visible Active States:** The current filters are explicitly stated in a "Time Range" badge at the top. Selected stations are highlighted with a distinct **accent color** and increased **marker size** to maintain user context.
- **Global Reset:** A "Reset All" button returns the system to the global overview state, ensuring a clear path for exploratory re-navigation.

Task 1: Compare Distributions (Member vs. Casual Trends)

The system enables users to identify behavioral differences between user groups. By observing the line chart, users can see that Members peak during commute hours (8 AM/5 PM), while Casual users show a smoother distribution throughout the day.

Task 2: Summarize Flows (OD Connectivity)

Using the curved arc map, the system reveals geographic patterns. Users can identify major routes (e.g., residential areas to transit hubs) through line thickness, with transparency handling high-density overlap.

Task 3: Rank Stations (Identify Top Hubs)

The bar chart reorders dynamically based on user filters. It helps planners identify which stations are the primary "sinks" or "sources" of bike traffic at any given time.

Task 4: Analyze Connectivity (1-Hop Neighborhoods)

When a station is selected, the map isolates its immediate network. This removes background noise and shows the specific "reach" of a station, helping to evaluate local network resilience.

Storyboard: Optimizing the Morning Commute

1. Global Overview: An urban planner opens the dashboard to see the full month's data. The initial view shows high density around Grove St and Newport.



2. Temporal Filtering: The planner notices a Member spike at 8 AM and brushes this range on the line chart. The map updates to show only rush-hour flows.



3. Identifying Hubs: The Bar Chart reorders, revealing "Grove St PATH" as the top destination. The planner clicks this bar to isolate the station on the map.



4. Discovering Insights: By isolating Grove St's 1-hop neighborhood, the planner identifies a heavy flow from residential Hamilton Park, but a lack of connectivity from newer developments, suggesting a site for a new station.

